

AI-Augmented Research

Session 1.2: Literature Discovery, Synthesis, and Research Design

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instats Seminar

Monday, August 10, 2026 7:30–9:00 pm

Interactive App and Repository

- **Streamlit App URL:** [Click here to open the Interactive Streamlit App](#)
- **No Installation:** Runs entirely in the browser via Streamlit — no Python or programming knowledge required.
- **Navigation:** Use the sidebar to switch between the six seminar sessions across the three days.
- **Step-by-Step:** Each page walks through worked examples and a Bring Your Own Data (BYOD) interface for that session's core workflows — from research question development and literature scoping to data quality auditing, end-to-end computational pipelines, manuscript integrity checks, and AI disclosure evaluation.
- **Consistency:** Results match the companion Python scripts and prompt library available in the `scripts/` folder of the seminar GitHub repository.
- **GitHub Page URL:** [Click here to open the seminar GitHub page](#)

Session 1.2 Overview

- 1 Discovery versus Verification in Literature Work
- 2 Citation Hallucination
- 3 AI-Assisted Literature Discovery
- 4 AI-Assisted Synthesis
- 5 Research Design and Methodological Planning
- 6 AI for Visual Communication
- 7 No-Code Workflow Demonstrations

The Discovery-versus-Verification Distinction in Literature Work

In literature work, the discovery-versus-verification distinction maps directly onto two phases of the research process.

Discovery phase

- Scoping a field
- Surfacing unexpected connections
- Identifying key authors and debates
- Proposing methodological approaches

Verification phase

- Checking every AI-generated citation against the primary source
- Confirming abstracts match summaries
- Verifying that cited claims appear in the cited paper

These two phases require different tools, different disciplines, and different documentation standards.

Citation Hallucination: The Most Consequential AI Failure Mode for Researchers

Citation hallucination — the generation of plausible-sounding but non-existent references — is **not a rare edge case**; it is a systematic property of current LLMs.

- Walters & Wilder (2023): **55%** of GPT-3.5 citations fabricated; **18%** of GPT-4 citations fabricated
- Naddaf & Quill (2026): tens of thousands of 2025 publications may contain invalid AI-generated references
- Chen (2026): hallucination rates of 14–95% across models in citation generation tasks

Non-negotiable rule

Every AI-generated citation must be verified against the primary source before use, without exception.

Why Hallucination Rates Vary So Widely

Key factors driving variation in citation hallucination rates:

- 1 **Domain specificity:** hallucination rates are higher in specialized domains (medical, legal) than in general domains
- 2 **Prompt structure:** open-ended citation requests produce higher hallucination rates than structured extraction from provided text
- 3 **Model generation:** newer models hallucinate less, but not negligibly
- 4 **Retrieval augmentation:** RAG systems grounded in a verified corpus dramatically reduce hallucination rates

Linardon et al. (2025): influence of topic familiarity and prompt specificity on citation hallucination.

Practical implication: RAG over a curated reading list is preferable to open-ended chat for literature work.

AI-Assisted Literature Discovery: The Search Mode

In Zyphur's (2026) taxonomy, literature discovery uses AI in **Search mode**: as a retrieval and scoping tool.

Appropriate tools for this mode are **retrieval-augmented systems** (Elicit, Consensus, Semantic Scholar, Perplexity with citations, NotebookLM) rather than general-purpose chat. These systems ground their outputs in indexed literature and provide source links, making verification tractable.

Key capabilities:

- Scoping a field
- Identifying key papers and authors
- Surfacing adjacent literatures
- Generating structured summaries

Key limitation:

- Coverage depends on the underlying index
- Grey literature and non-English sources are systematically underrepresented

Retrieval-Augmented Generation (RAG) for Literature Work

RAG systems combine a **retrieval component** (searching a corpus) with a **generation component** (producing a grounded response). For literature work, RAG over a curated reading list is preferable to open-ended chat because:

- ① Outputs are grounded in verifiable sources
- ② Hallucination rates are dramatically lower
- ③ The researcher controls the scope of the corpus
- ④ Source attribution is tractable

Practical tools:

- **NotebookLM**: upload PDFs, query against them
- **Elicit**: structured literature extraction with citations
- **Consensus**: evidence synthesis with citations

AI-Assisted Synthesis: Opportunities and Limits

AI can assist with:

- Identifying thematic clusters across a corpus
- Extracting structured data from papers
- Generating structured summaries
- Proposing a narrative organization

AI cannot:

- Evaluate the quality of the evidence
- Resolve genuine theoretical disagreements
- Determine what the synthesis means for the field

The researcher's intellectual authority is most critical at the synthesis stage. Mtotywa et al. (2026): a capability-requirement matrix should map AI tool capabilities to specific synthesis tasks.

Prompt Discipline in Literature Discovery

The choice of search query or synthesis prompt is a **researcher degree of freedom**. Documentation standard for AI-assisted literature work:

- 1 Record the exact query or prompt
- 2 Record the tool, version, and date
- 3 Record the number of results and the selection criteria applied
- 4 Verify a random sample of citations against primary sources
- 5 Report the verification rate and any errors found

This is the minimum required for a reader to assess the validity of the AI-assisted literature review.

Research Design and Methodological Planning with AI

AI can assist with research design by:

- Identifying methodological precedents in the literature
- Explaining the assumptions of unfamiliar methods
- Comparing design options
- Identifying potential threats to validity

The **Tutor mode** (Zyphur 2026) is appropriate here: corpus-grounded retrieval over methodological literature is preferable to open-ended chat.

Key verification discipline: the researcher must be able to explain and defend every methodological choice independently. AI assistance does not transfer methodological authority.

AI for Visual Communication: An Underused Research Tool

Researchers spend substantial time on visual outputs — figures, diagrams, conceptual illustrations, presentation slides, and posters. AI tools now make high-quality visual communication accessible without specialist design skills.

A figure that misrepresents the structure of an argument or the design of a study is a validity problem, not merely an aesthetic one.

Visual type	Research function	AI tool examples	Disclosure?
Conceptual diagrams	Represent theoretical models	Napkin.ai, Whimsical AI	Yes, per journal policy
Data figures	Communicate empirical results	Julius AI, ChartAI	Depends on involvement
Workflow diagrams	Document research process	Lucidchart AI, Mermaid	Usually not required
Presentation slides	Communicate to audiences	Gamma, Beautiful.ai	Usually not required
Posters	Conference communication	Canva AI	Usually not required

Chen (2026): AI-generated figures in academic publishing — policies, tools, and disclosure requirements.

Demonstration: Literature Discovery with Elicit or Consensus

Step-by-step workflow

- 1 Formulate the research question in plain language
- 2 Enter the question into the tool's search interface
- 3 Review the structured results (titles, abstracts, extracted data)
- 4 Identify the 5–10 most relevant papers
- 5 **Verify each citation:** copy the DOI or title into Google Scholar or PubMed and confirm the paper exists and the abstract matches
- 6 Export the verified corpus for synthesis

Verification step: of 15 AI-generated citations, expect 10–13 to be usable after verification. This is a typical outcome and illustrates why verification is non-negotiable.

Demonstration: Synthesis with NotebookLM

Step-by-step workflow

- 1 Upload 10–20 verified PDFs to a NotebookLM notebook
- 2 Ask a synthesis question: “What are the main methodological approaches used in this literature?”
- 3 Review the response and the source citations provided
- 4 Click each source citation to verify it against the uploaded PDF
- 5 Ask follow-up questions to explore specific themes
- 6 Export the synthesis notes for use in the literature review

Key advantage of RAG: every claim in the response is linked to a specific passage in a specific uploaded document, making verification tractable.

Demonstration: AI as Cognitive Partner for Research Design

Step-by-step workflow

- 1 State the research question and proposed design in plain language
- 2 Ask: “What are the main threats to internal validity in this design?”
- 3 Evaluate each threat: is it substantive or superficial?
- 4 Ask: “What is the strongest argument against this design?”
- 5 Apply **model-heterogeneity discipline**: repeat steps 2–4 with a different AI system
- 6 Document both sets of responses and the researcher’s evaluation of each

Zyphur (2026) Competency 4: persistent critique across AI families is stronger evidence of a genuine problem than single-model critique.

Demonstration: AI for Visual Communication

Step-by-step workflow

- 1 Describe the theoretical model or research design in plain language
- 2 Use Napkin.ai or Whimsical AI to generate a first-draft diagram
- 3 Evaluate: does the diagram accurately represent the structure of the argument?
- 4 Iterate: provide corrections in plain language and regenerate
- 5 Export the final diagram and document the AI tool used
- 6 Check the target journal's policy on AI-assisted figures

Chen (2026): disclosure is warranted whenever an AI tool materially shapes the research output. For figures generated from scratch by AI, disclose the tool, version, and the description provided.

The Prompt-as-Fork-in-the-Garden: A Demonstration

The same research question, three different prompts, three different literature scopes:

- **Prompt A:** “What is the literature on X?” → broad, unfocused results
- **Prompt B:** “What are the most-cited empirical studies on X published since 2020?” → more focused, but recency bias
- **Prompt C:** “What are the main methodological debates in the literature on X?” → methodologically oriented, different papers

The prompt is a methodological choice. Document it. (Zyphur 2026, Competency 3)

Session 1.2 Summary

Three things to carry forward

- 1 **Citation hallucination is systematic, not rare:** every AI-generated citation must be verified against the primary source before use, without exception
- 2 **RAG over a curated corpus is preferable to open-ended chat** for literature work: it grounds outputs in verifiable sources and makes verification tractable
- 3 **Prompt discipline applies to literature discovery:** the choice of search query or synthesis prompt is a researcher degree of freedom that must be documented

References

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Questions and Discussion

Thank you!

Questions?

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